

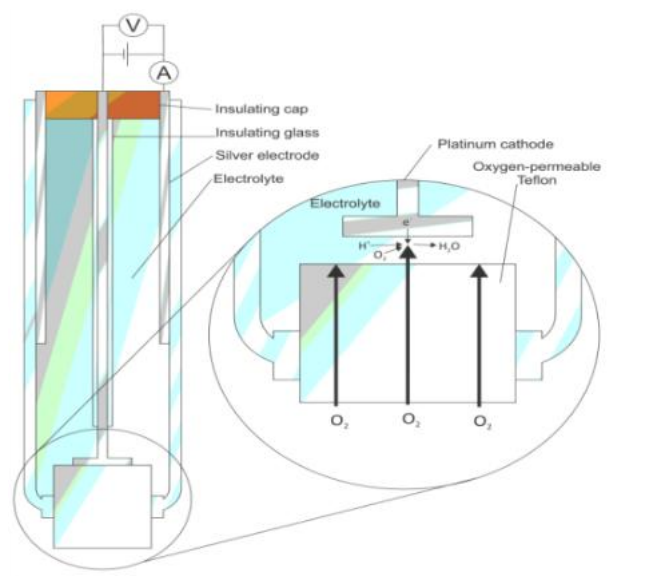
## Electrochemical Nano-Biosensors

### Introduction

Biosensors, with their wide range of applications from disease monitoring to water and food control, are an important supporter of today's quality of life. They have played a key role, especially in rapid diagnosis (POC) technologies (such as blood pressure monitors, pregnancy tests, and COVID-19 tests). The key to their success is the portability and low cost they provide with their selective biorecognition capabilities. Among these platforms, electrochemical biosensors (speed, low cost, and smartphone integration) stand out. This interdisciplinary field, which extends from materials science to medicine, began its journey with the Clark oxygen electrode and is now progressing towards the future of nanobiosensors (from microbe detection to cancer diagnosis) and artificial intelligence-supported smart platforms.

### 1. Basic Properties of Biosensors

In recent years, the demand for chemical sensors and biosensors has increased rapidly due to serious challenges in the environmental and health fields [1]. While traditional chemical analysis methods offer advantages such as high accuracy and acceptable sensitivity, they often require complex and costly instruments and require excessive reagent and sample consumption. Furthermore, their lack of portability for on-site monitoring and the need for highly qualified laboratory personnel limit their widespread use [2]. These limitations have necessitated the development of new chemical sensor and biosensor platforms. The rapid development of biosensor technology began in 1962 after Clark and Lyons introduced the first glucose oxidase biosensor (an electrode that measures the partial pressure of ambient oxygen in a liquid with a catalytic platinum surface), as shown in **Figure 1** [3].



**Figure 1:** Schematic depiction of Clark and Lyons' oxygen electrode

Since then, numerous sensor and biosensor applications have been described, and some have been commercialized. A biosensor, in its simplest form, is an analytical device in which a biologically active element is in close contact with a suitable physicochemical transducer, producing a measurable signal (optical, electrical, or electrochemical) directly proportional to the concentration of the target substance [4, 14].

Biosensors are flexible and highly valuable detection techniques that have the potential to solve problems and challenges in various fields such as animal safety, drug and pharmaceutical analysis, environmental monitoring, food safety, and defense [5,6]. One of their most important advantages is the ability to directly examine target analytes in often complex samples without the need for preprocessing [6]. Additionally, features such as simultaneous multiple target recognition with high selectivity, full automation, and reduced cost and sample size can be achieved [5, 6]. The most common example in daily life is the glucometer, a blood sugar monitoring device shown in **Figure 2** [7, 12, 16].

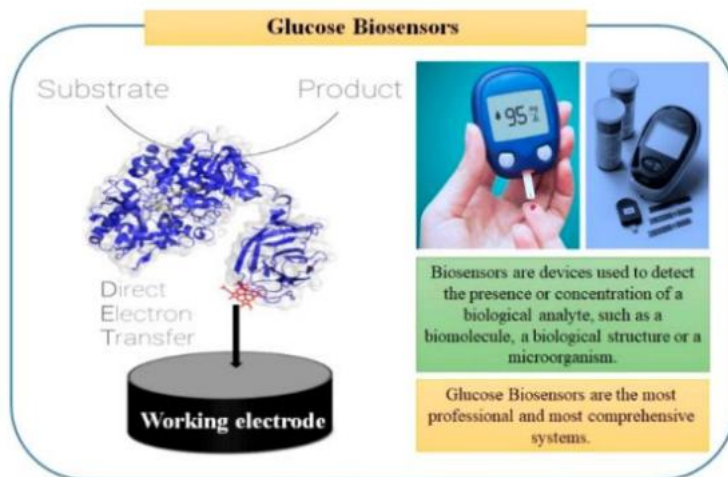
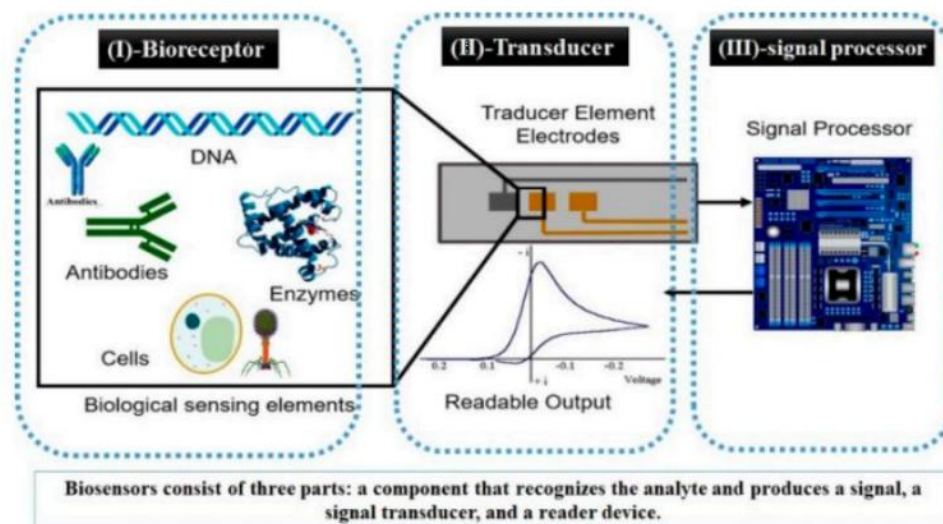


Figure 2. Glucose biosensors

A typical biosensor consists of three main components as shown in **Figure 3**, biorecognition element (bioreceptor), transducer, and signal processor.



**Figure 3:** A common diagram showing the main building blocks of biosensors.

**Biorecognition Element (Bioreceptor):** Designed for high-specific recognition of a selected analyte, it is the key component that critically regulates the biosensor's high selectivity and sensitivity.

**Transducer:** Directly connected to the bioreceptor and responsible for converting biological/biochemical responses into measurable signals.

Signal Processor: Amplifies small signals from the transducer into larger output signals containing important waveform characteristics. The digital signal processor collects, analyzes, displays, and stores this amplified data in the device's internal memory [6].

## **2. Brief History of Biosensors**

The history of biosensors began with Leland Charles Clark's first report on a blood oxygen sensor in 1956 and solidified with the 1962 experiment with enzyme-based glucose sensing [3 13, 16]. The first generation (mediatorless) sensors measured the byproduct of a bioreaction, and the second generation (mediator) sensors (1976) used artificial redox mediators to facilitate electron transfer and improve detection limits. Later, with the third generation, portability became a priority. Today, biosensor technology has reached the nanotechnology era [8], where nanomaterials are used for superior performance.

## **3. Biomedical and Environmental Applications**

Electrochemical biosensors are widely used in a wide range of applications such as microbial detection, cancer diagnosis, and food quality control, as microbial infections, for which early detection is crucial, are a global challenge [5,6]. The most frequently used platforms for pathogen detection are electrochemical immunosensors that immobilize a biorecognition element to the electrode surface [8, 15]. To increase selectivity, structures such as Molecular Imprinted Polymers (MIPs), which are chemically generated in the presence of the target molecule or cell (for the detection of *Salmonella typhimurium*, and bacteriophage-based sensors that exhibit high specificity and stability to the host bacteria (for *Listeria innocua* have also been introduced [1,9]. These advanced methods enable sensitive and rapid microbial diagnosis by providing signal amplification with the help of nano-electrochemical transducers [10, 11, 17].

## **4. Conclusions, Future Remarks, and Perspectives**

Beginning with Clark and Lyons's first glucose biosensor, biosensors, with their selective recognition, low cost, and portability, are the cornerstone of today's rapid diagnostic (POC) technologies [4,6]. Electrochemical biosensors, offering both low cost and high speed, are particularly prominent. The integration of nanotechnology has exponentially increased the sensitivity and selectivity of sensors thanks to nanoscale materials, enabling nanobiosensors to perform rapid and accurate analyses in a wide range of applications, from microbial detection to cancer diagnosis. The global biosensors market is expected to reach \$26 billion by 2025, and this growth is fueled by increasing demand for portable devices due to factors such as diabetes [1]. As a result, biosensors are evolving into intelligent platforms powered by AI and machine learning to deliver groundbreaking proactive solutions in personalized medicine and early diagnosis.

## **References:**

[1] Hassan, R. Y. A. (2022). Advances in Electrochemical Nano-Biosensors for Biomedical and Environmental Applications: From Current Work to Future Perspectives. *Sensors*, 22(19), 7539.

- [2] Schoephoerster, J., Jha, N., Singh, A., Khara, A., & Awasthi, S. (2022). Nanomaterials-Based Electrochemical Biosensors: Advancing the Rapid and Sensitive Detection of Bacterial Pathogens in Food Samples. *Sensors*, 22(7), 2689.
- [3] Clark, L. C., & Lyons, C. (1962). Electrode systems for continuous monitoring in cardiovascular surgery. *Annals of the New York Academy of Sciences*, 102(1), 29–45.
- [4] Thevenot, D. R., Toth, K., Durst, R. A., & Wilson, G. S. (2001). Electrochemical biosensors: Recommended definitions and classification. *Pure and Applied Chemistry*, 73(2), 163–182.
- [5] Dzyadevych, S. V., Arkhypova, V. M., & Soldatkin, O. P. (2008). Amperometric biosensors for environmental monitoring. In *Biosensors for Environment, Health and Food Safety* (pp. 119–152). Springer.
- [6] Grieshaber, D., Lingenfelter, J., Peveler, D., Davis, J., Rylander, M., & Tothill, I. (2008). Electrochemical biosensors: Sensor principles, concepts and applications. *Sensors*, 8(3), 1400–1457.
- [7] Yañez-Sedeno, L., Alonso-Lomillo, M. A., Toca-Herrera, J. L., & Parrado, M. S. (2021). Electrochemical biosensors for clinical applications: An overview. *Electrochemistry Communications*, 132, 107142.
- [8] Bhalla, V., Vasta, B., & Vasta, P. (2016). Biosensors. In *Biosensors and Bioelectronics* (pp. 1–28). Springer.
- [9] Turner, A. P. F., & Piletsky, S. A. (2001). Molecularly imprinted polymers in biosensors. *Pure and Applied Chemistry*, 73(12), 1443–1448.
- [10] Zhang, M., Li, X., Pan, J., Zhang, Y., Zhang, L., Wang, C., Yan, X., Liu, X., & Lu, G. (2021). Ultrasensitive detection of SARS-CoV-2 spike protein in untreated saliva using SERS-based biosensor. *Biosensors and Bioelectronics*, 190, 113421.
- [11] Choi, J. R., Choi, M., Han, E. T., Chung, S., Kim, J., Kim, H., Kim, M., & Lee, N. (2020). A review of smartphone-based biosensors for infectious diseases: Toward practical point-of-care diagnosis. *Biosensors and Bioelectronics*, 170, 112675.
- [12] Guilbault, G. G., & Montalvo, J. G. (1969). Electrode for the ultra-fast estimation of urea in blood and urine. *Analytical Letters*, 2(5), 283–298.
- [13] Higgins, I. J., Scott, D., & Scott, B. R. (1984). The enzyme electrode: An innovative new biosensor. *Analytical Chemistry*, 56(13), 1827A–1831A.
- [14] Liedberg, B., Nylander, C., & Lundstrom, I. (1983). Surface plasmon resonance for gas detection and biosensing. *Sensors and Actuators*, 4, 299–304.
- [15] Lupfer, E., & Hanke, L. (2022). A comprehensive review of biosensor technology with applications in the food industry. *Biosensors and Bioelectronics*, 198, 112814.
- [16] Updike, S. J., & Hicks, G. P. (1967). The enzyme electrode. A new approach to electrochemical sensors. *Nature*, 214(5092), 986–988.
- [17] Wang, S., Kim, D., Lim, J., Yang, C., & Lee, H. (2022). A review on nanoparticle-based biosensors for infectious disease diagnosis. *Biosensors and Bioelectronics*, 202, 114175.

